Introduction

Course goals and some fundamentals

Introduction

The components of complex information technologies solutions

Software

Databases, web technologies, programs, networking technologies

Hardware

Server computers, end user computers (PCs, notebooks, smartphones), networking hardware, input/output devices (e.g., sensors, scanners), ...

People

Users, system administrators, deverlopers, support staff, ...

Any solution requires components from all three catagories

We will focus on the software components

Introduction

Two common approaches

Focus on the individual components, work with small examples, build understanding bottom-up, combine later

or

Look at the whole solution, the bigger picture, try to understand how components work together



This is what we are doing in this course by looking at Moodle

The hope is that you

Get inspired to learn more about the building blocks

Gain confidence in your ability to understand complex systems

Find it interesting to analyse a real life system used by millions of users around the world

Introduction

What is Moodle?

Moodle is a Learning Management System (LMS)

https://moodle.org/

LMS

- Are used by educational institutions (mainly tertiary and secondary) to support teaching and learning
- They build on class/course structures to group learners who are supported by teachers
- Common uses are to make material available, to facilitate interaction and assessment

Sounds familiar?

- Massey's Stream is based on Moodle

Introduction

What is Moodle conceptually?

Moodle is a Learning Management System (LMS)

What is Moodle technically?

Moodle is a web-based system

It provides its users access via a web interface

Behind the scenes, it uses web and networking technologies, databases storage, programming languages, ...

Introduction

Why do we focus on Moodle in our course?

You are familiar with the concepts of a LMS and with Moodle in its installation as Stream

Moodle has all the components of a typical complex information technology solution

Moodle uses common tools for its components (e.g., PHP, Apache, MySQL)

Moodle is open source, has open development processes and an active user/support community

Moodle is used world-wide in about 250 countries, has about 160,000 registered sites and more than 2 billion users (see https://stats.moodle.org/)

Take time to reflect on how what we learn in the context of Moodle transfers to more general contexts

Introduction

The difference between Moodle and a Moodle site

- At moodle.org you find ‘Moodle’
  - You find information about Moodle
  - You find the Moodle code to download
  - If you have the right hardware (a server or a PC) you can create your own Moodle site
    - You will be in charge of your own Moodle site
    - You can decide what courses you setup, which learners you enrol, who has access, ...
- As Moodle is open source, anyone can create their own Moodle site
- The value of a Moodle site depends on its organizational context
  - What are the differences between Stream (Massey’s Moodle site) and a Moodle site you might create?

Introduction

Your own Moodle sites for this course

For the purposes of 158.120 everyone gets their own Moodle site

Why?

- This puts you in charge of your site
- You are the administrator, you decide which data you define
- This allows you to follow through how these data are stored and processed

We could have asked you to install your own sites ...

- Installing a site should be easy in principle, but in reality often produces challenges

... we have decided to give you sites installed on one of our servers

Setting up data for your Moodle site
is your first tasks hands-on task

Introduction

What is our focus?

We are interested in exploring Moodle from a technical perspective

- How are the Moodle webpages created?
- Where and how are data stored?
- How do data travel across the network?

You need to learn a little bit on using Moodle as administrator and teacher

- How do we setup a course? How do we create users?
- We need data to work with

In limited ways, you will visit each others Moodle sites

- You will log in to your own site as administrator, as teacher, as student
- You will invite others to access your site as teacher or student

Introduction

The people involved – with a Moodle site

Typically, a Moodle site is setup to support teaching and learning for an institution

This implies a hierarchy

- The institution controls the Moodle site
- The institution decides who has what type of access

We have a hierarchy of users

- Admin, teacher, teaching assistant, student

'Users' use Moodle via the Moodle web interface

IT specialists at the institution create and maintain the Moodle site

- Web server admin, network admin, database admin
- These individuals do not use the Moodle web interface

Introduction

The people involved – with Moodle

A Moodle site requires the installation of the Moodle software

Who provides the Moodle software?

Software developers: programmers, database specialists, web systems specialists, user interface specialists, requirements analysts, software testers, translators, ...

Where do these people come from?

The key group belongs to Moodle HQ (head quarters), the group around the Moodle founder Martin Dougiamas

Others come from the Moodle open source community: From Moodle Partners, educational institutions, individuals

Important to distinguish the people who provide Moodle

From the people who maintain a specific Moodle site

From the people who use a Moodle site

Introduction

Which group of people will you represent in this course?

Software developers (who provide Moodle code)

- In a very limited way

IT specialists (who provide and maintain a Moodle site)

- No, you are given a ready installed Moodle site
- Yes, we will look at the database, the Moodle files (configuration files and program code) directly without using the Moodle web interface

Users (who work with Moodle via a the Moodle web interface)

- As Moodle admin (who sets up courses, users, enrolments)
- As students and teachers to ‘use’ Moodle

Introduction

What is the role of the Moodle web interface?

Users (admin, teachers, students) access a Moodle site via the Moodle web interface

- The user points their web browser to the Moodle site (e.g., stream.massey.ac.nz)
- The user provides their credentials (username and password)
- Based on the credentials Moodle controls what the user can see and do

Web server, file system and database settings prevent users from accessing Moodle files and user data directly

- This is crucial to the functioning of a system like Moodle!

The Moodle admin is a user who is restricted to the web interface

The IT specialist accesses Moodle data and programs without restrictions imposed by the web interface

Introduction

Review questions

What are the components of complex information technology solutions?

How do the following terms relate to each other?

LMS, Moodle, open source, site, Stream

What groups of people are involved with Moodle and a Moodle site?

What are the roles of these people, what do they focus on, what do they need to know?

Introduction

Where to from here

Browse through Part 1 of the recommended readings textbook

Take a quick look at moodle.org

Look at the next presentations in this course section

Work on your own Moodle site